TCGA-BRCA: Gene Expression and PAM50 Visualization

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Introduction

Breast cancer represents a major global public health challenge requiring multinational collaboration in scientific research and pharmaceutical development. In 2024, approximately 2.4 million women were diagnosed with breast cancer worldwide, with an estimated 694,000 associated deaths (World Health Organization, 2026). Despite advances in screening and treatment, breast cancer remains a biologically heterogeneous disease, exhibiting considerable variation in molecular characteristics and clinical behavior.

Through the contributions of scientists worldwide, gene-expression research has identified distinct molecular subtypes, including Luminal A, Luminal B, HER2-enriched, and Basal-like breast cancers (Parker et al., 2009). These classifications demonstrate differences in tumor biology and provide researchers with a framework for investigating variation in gene expression. The Cancer Genome Atlas (TCGA) has established an extensive molecular resource for investigating breast cancer heterogeneity (Cancer Genome Atlas Network, 2012).

Two genes, ESR1 and ERBB2, were selected for their associations with estrogen signaling and HER2-related growth pathways, respectively (Cancer Genome Atlas Network, 2012). ESR1 (Estrogen Receptor 1) encodes estrogen receptor alpha, which regulates hormone-dependent gene expression and is strongly associated with luminal breast cancer subtypes (Parker et al., 2009). ERBB2 (HER2) encodes a receptor tyrosine kinase involved in cellular growth and survival, with amplification or overexpression associated with HER2-positive breast cancer (Cancer Genome Atlas Network, 2012).

The Data

The research examined a gene-expression count matrix (`counts.csv`) and associated sample metadata (`meta_data.csv`) provided through the Foundations in Data Science course

(HSE.711) at the Geisel School of Medicine at Dartmouth.

The files were attributed to The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) project. The results presented here are based, in whole or in part, on data attributed to the TCGA Research Network. Although the supplied data closely resemble TCGA-BRCA records, the original acquisition methods, processing procedures, and provenance of the course-distributed files have not been independently verified. Consequently, the precise relationship between the supplied files and the original TCGA-BRCA release remains pending independent validation.

The dataset contained 60,660 gene-level measurements across 1,231 samples, accompanied by sample annotations.

TCGA was made possible through the voluntary contributions of patients who consented to the use of their biological specimens and associated clinical information for cancer research. These contributions enabled substantial advances in the genomic and molecular characterization of cancer. TCGA established informed-consent procedures and data-access safeguards to protect participants while supporting scientific research. The contributions of these patients continue to advance our understanding of breast cancer heterogeneity.

Methods

Data were loaded and analyzed using R version 4.4.3 (aarch64-apple-darwin20) on macOS 26.5.2, through the RStudio development environment. Data preparation and manipulation employed the `data.table`, `dplyr`, `tibble`, and `stringr` packages. Descriptive statistics and statistical comparisons were performed using base R functions.

Visualizations were developed using `ggplot2`, with `ComplexHeatmap` and `circlize` supporting gene-expression heatmaps and standardized color mappings. Course lectures and the R Graph Gallery served as instructional resources during heatmap development. Summary tables and manuscript outputs were prepared using `knitr` and `kableExtra`. Final figures and tables were exported for incorporation into this LaTeX manuscript.

Sample identifiers were aligned by matching count-matrix column names to the metadata `barcode` field. The analysis included primary tumor, solid tissue normal, and metastatic samples. PAM50 subtype comparisons were restricted to primary tumors with classifications recorded in the `paper_BRCA_Subtype_PAM50` field. Of 1,111 primary tumor samples, 1,099 had available subtype annotations; 12 were excluded from subtype-specific analyses because their classifications were missing.

For exploratory visualization, ESR1 and ERBB2 raw-count measurements were transformed using $\log_2(x+1)$ or $\log_{10}(x+1)$, depending on the graphical display. Ten breast cancer-associated genes were selected for heatmap visualization: ESR1, PGR, GATA3, FOXA1, ERBB2, GRB7, EGFR, KRT5, KRT14, and MKI67. Gene-expression measurements were standardized using row-wise Z-scores to illustrate relative expression patterns across samples. Z-score tails were used as descriptive indicators of expression variability rather than measures of statistical significance.

Descriptive statistics, including means, medians, standard deviations, and interquartile ranges, were calculated to characterize gene-expression distributions. Linear regression was used to explore the relationship between ESR1 and ERBB2 log-transformed counts.

An exploratory volcano plot compared primary tumor and solid tissue normal samples. Gene-level counts were converted to counts per million (CPM), and log2 ratios of group mean CPM values were calculated using a pseudocount of one. Welch's t-tests were used for exploratory comparisons, with resulting p-values adjusted using the Benjamini–Hochberg false-discovery-rate procedure (Benjamini & Hochberg, 1995).

The volcano plot illustrated a log2 ratio threshold of ± 1 and an adjusted p-value threshold of 0.05. Because this analysis did not account for sample pairing, batch effects, or RNA-seq count dispersion using a suitable count-based model, the results were interpreted as **exploratory rather than confirmatory** (Love et al., 2014).

Results

[Report the final matched sample count and molecular subtype distribution, including denominators and missing annotations.]

[Report reproducible descriptive findings from the gene-expression summaries.]

[Reference the final heatmap and briefly describe its most relevant expression patterns, noting that Z-score tails do not represent statistical significance.]

[Reference the final volcano plot and identify the comparison groups, axes, thresholds, and methodological limitations. Do not interpret exploratory highlights as validated differential-expression findings.]

Figures

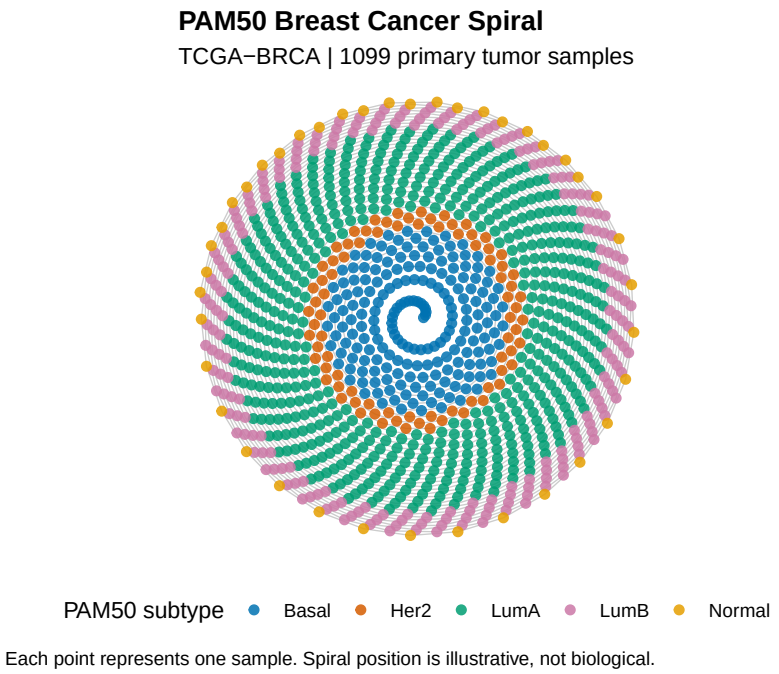


Figure 1: Visualization of PAM50 molecular subtypes in the TCGA-BRCA cohort.

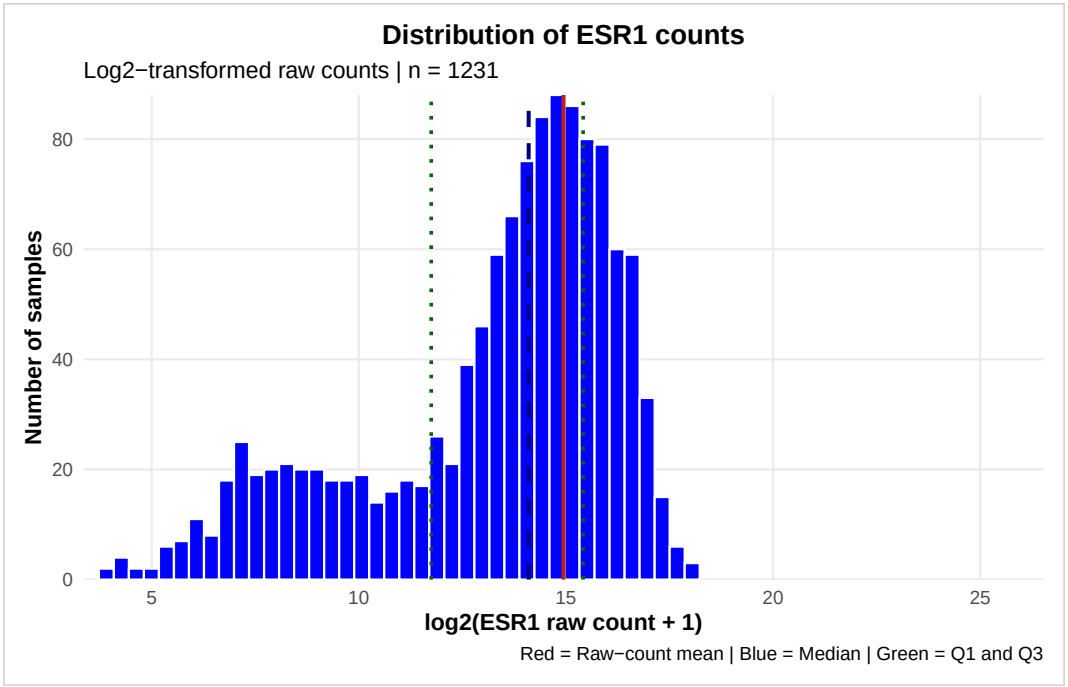


Figure 2: Distribution of ESR1 gene-expression values.

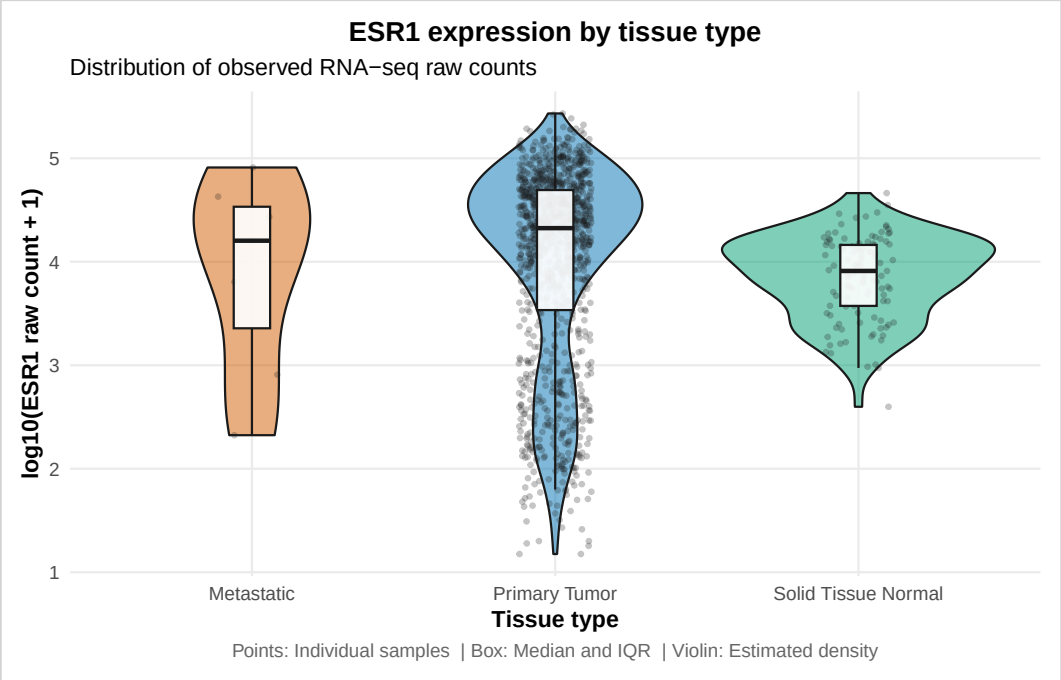


Figure 3: ESR1 expression across sample types.

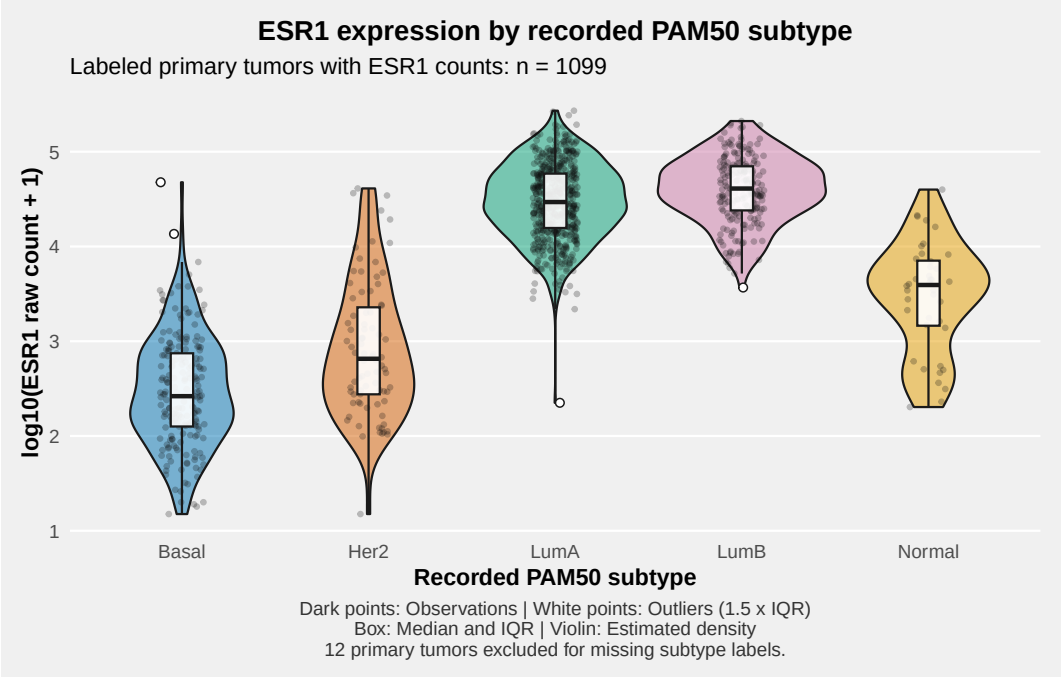


Figure 4: ESR1 expression across PAM50 molecular subtypes.

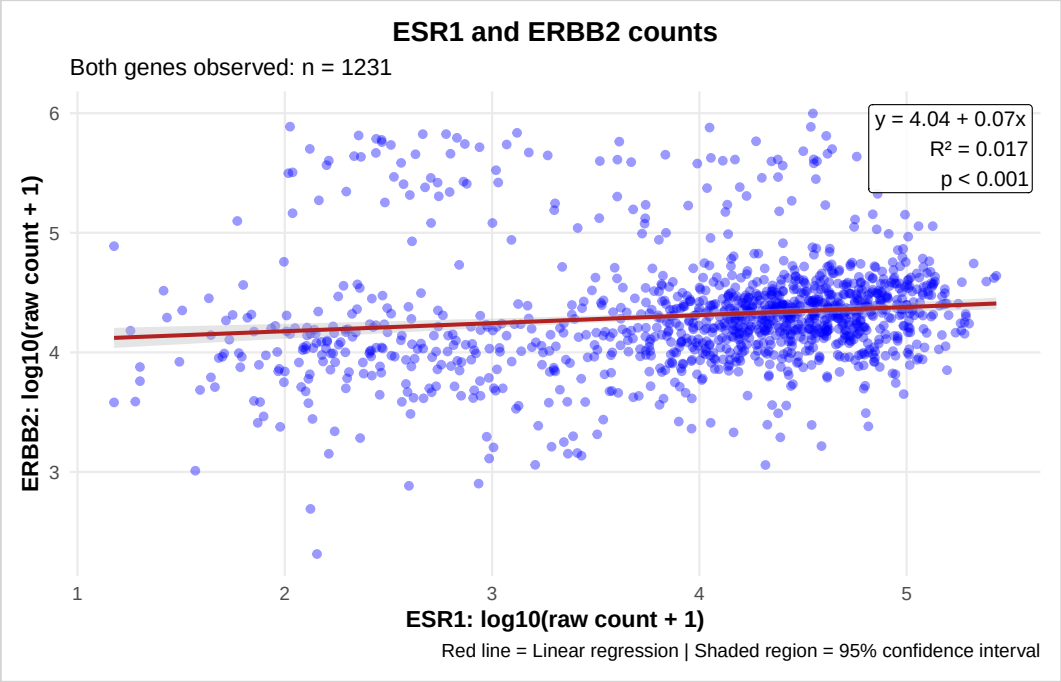


Figure 5: Exploratory gene-expression scatterplot.

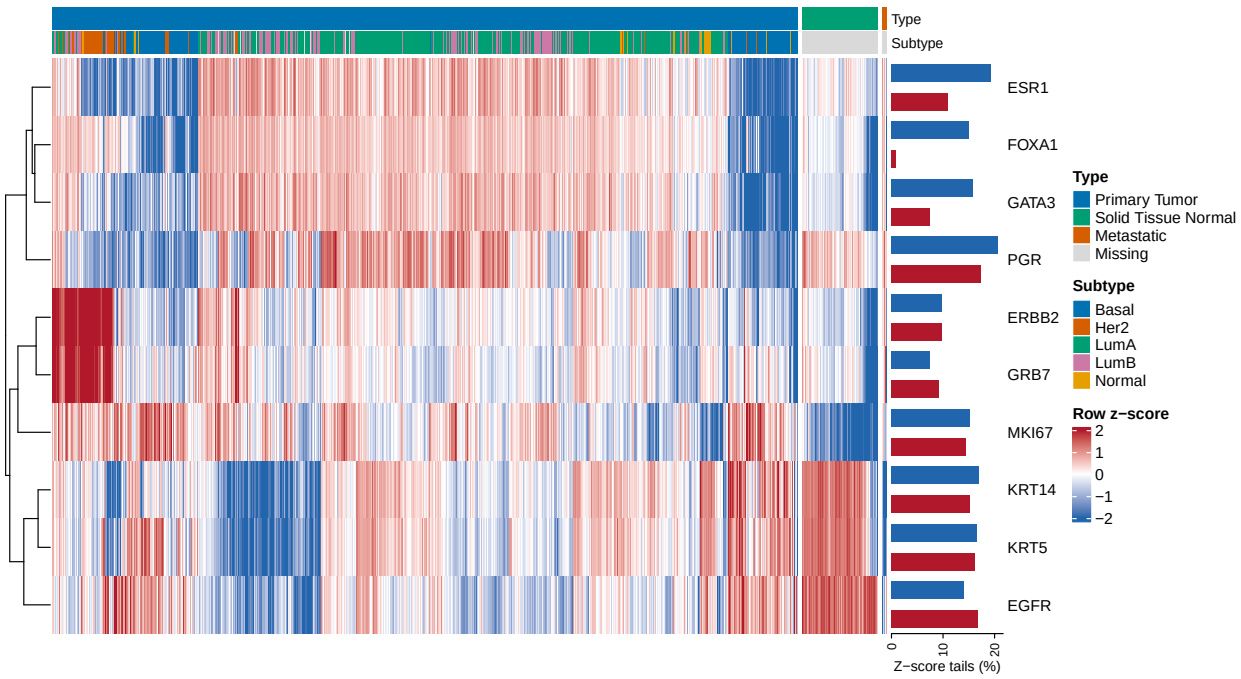


Figure 6: Heatmap of selected breast cancer-associated gene-expression patterns.

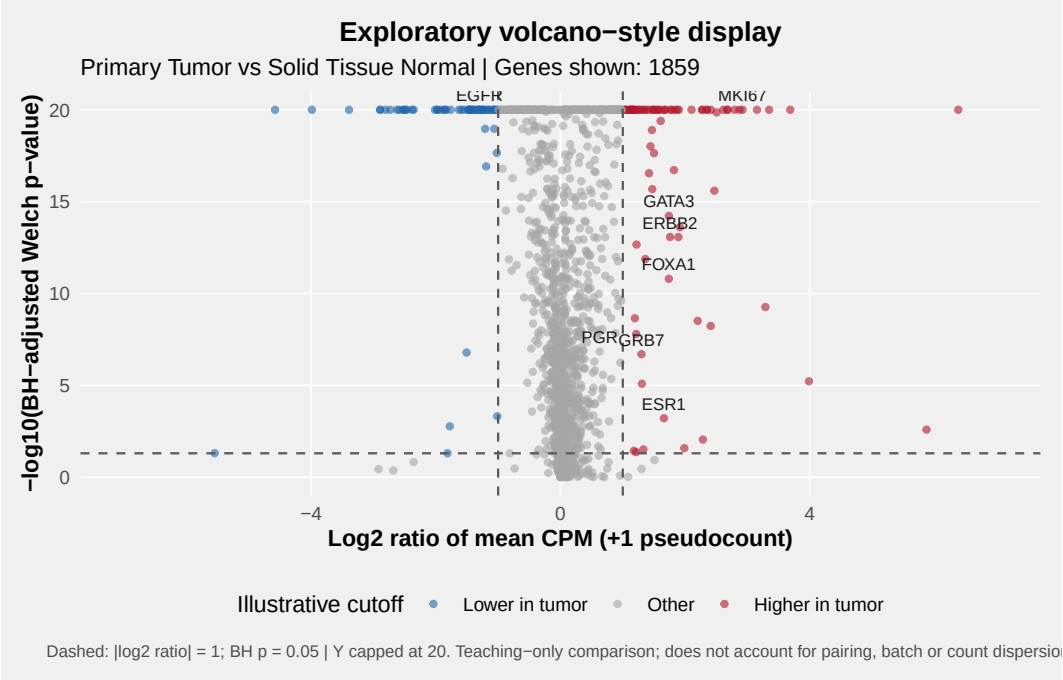


Figure 7: Exploratory volcano plot of gene-expression differences between comparison groups.

Tables

Table 1: Primary tumors with a recorded PAM50 subtype ($n = 1{,}099$).

Subtype	Samples	Percent (%)
Basal	197	17.9
Her2	82	7.5
LumA	571	52.0
LumB	209	19.0
Normal	40	3.6

Table 2: ESR1 and ERBB2 raw-count descriptive statistics.

Gene	N	Missing	Mean	Median	SD	IQR	Min	Max
ESR1	1,231	0	31,442.2	17,561	38,261.2	40,242.5	14	270,977
ERBB2	1,231	0	46,982.5	19,125	105,287.5	19,287.0	205	995,067

Table 3: ESR1 raw-count statistics by PAM50 molecular subtype.

Subtype	N	Mean	Median	SD	IQR	Min	Max
Basal	197	924.5	262.0	3,589.1	620.0	14	47,589
Her2	82	3,414.2	652.5	7,745.6	2,010.8	14	40,971
LumA	571	43,811.7	29,436.0	39,799.2	43,034.5	223	270,977
LumB	209	53,197.4	40,849.0	41,119.8	46,128.0	3,686	210,656
Normal	40	6,066.0	3,922.5	7,775.7	5,626.5	201	39,845

AI Disclosure

AI was used to type in the file names for the figures. The main file before modifications was cleaned by AI from an old manuscript to empty the file, and pre-cook with reporting basic instructions - NO AI USED FOR MANUAL TYPE UP. After template was cooked, this was hand written, and typed with the exception of the file names. VIEW GITHUB for documentation chain.

Note: LaTeX beautify applied where noted. Beautify is not classified as an AI tool. I understand this tool to be a regressor or neural classifier tool. It cleans up the presentation of the laTeX.

In accordance with the course AI policy, the room_237 repository is maintained manually. Repository operations are performed through the macOS Terminal or GitHub Desktop. The manuscript and associated research files are edited and organized locally.

References

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, *57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Cancer Genome Atlas Network. (2012). Comprehensive molecular portraits of human breast tumours. *Nature*, *490*(7418), 61–70. https://doi.org/10.1038/nature11412

Love, M. I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for rna-seq data with DESeq2. *Genome Biology*, *15*(12), 550. https://doi.org/10.1186/s13059-014-0550-8

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